Mood Analysis Report

User Mood Monitoring System — MoodLens

1. User Information

User ID	moingardi10@gmail.com
Report Period	2026-03-07 to 2026-03-07
Total Entries	4
Model Used	XGBoost (primary) + Random Forest (debate) — 2/4 weeks unanimous
Questionnaire Version	Unified Mood Scale (30 items)

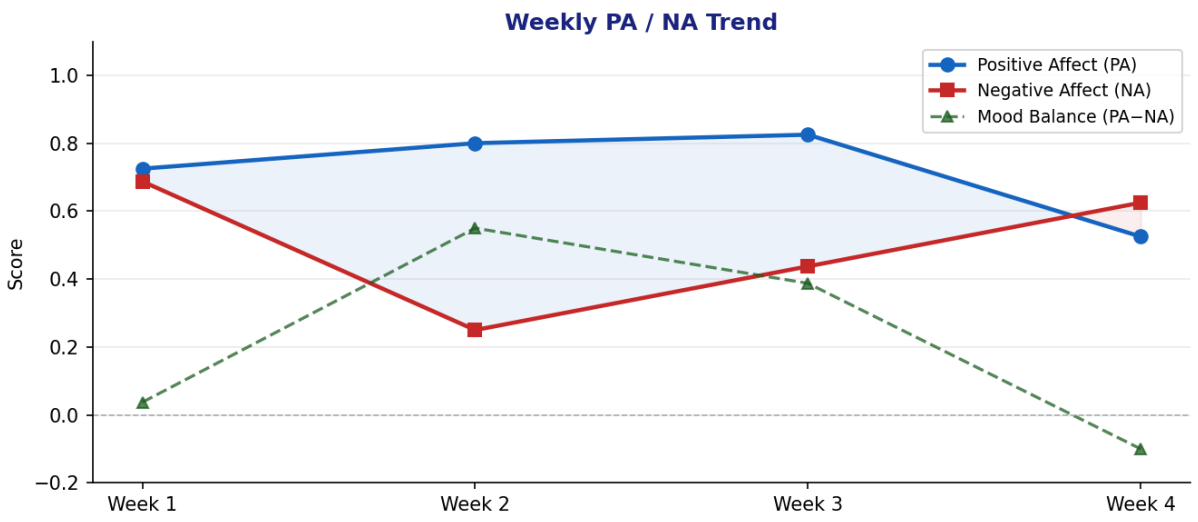
2. Overall Mood Summary

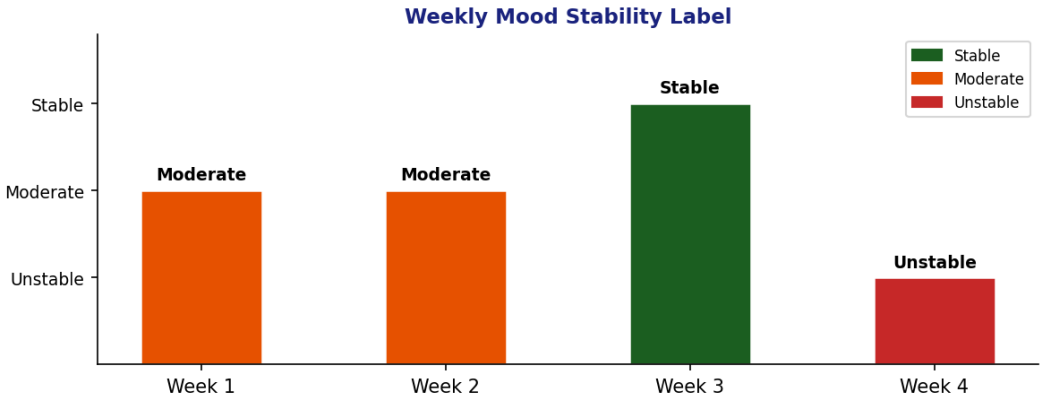
0.719	0.500	+0.219
Avg Positive Affect (PA)	Avg Negative Affect (NA)	Mood Balance (PA – NA)

Interpretation: Positive balance — overall positive mood state. | Overall Trend: Declining

3. Weekly Mood Trend

Week	PA Score	NA Score	Balance	Label
Week 1	0.725	0.688	+0.037	Moderate
Week 2	0.800	0.250	+0.550	Moderate
Week 3	0.825	0.438	+0.388	Stable
Week 4	0.525	0.625	-0.100	Unstable

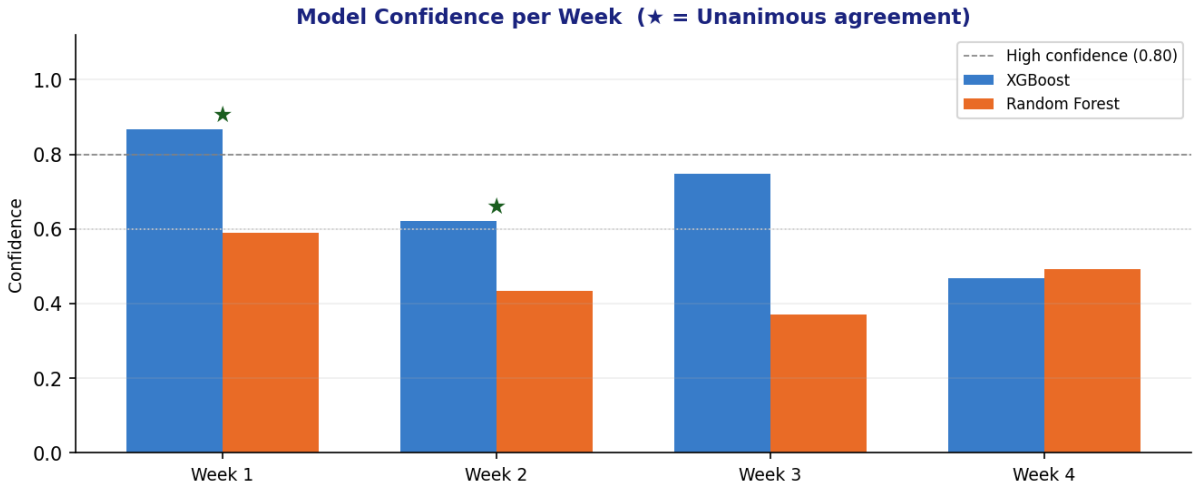


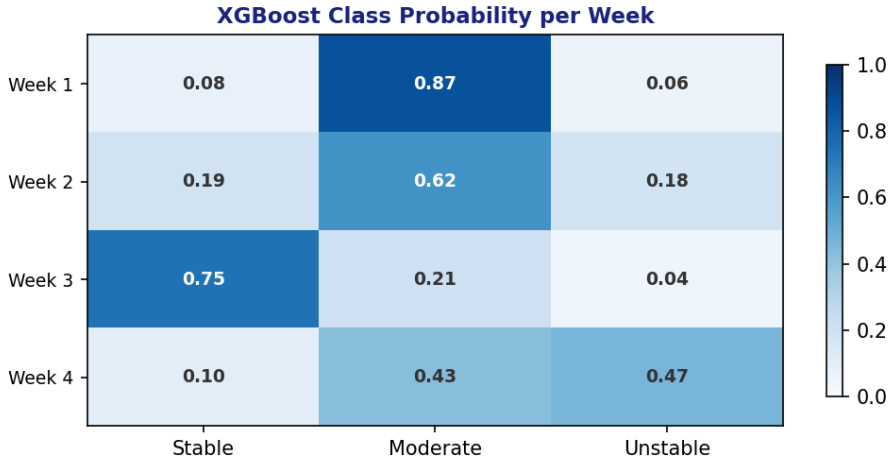


Model Debate Results

Each week, XGBoost and Random Forest independently predict the stability label. The debate engine selects the final label based on agreement, confidence gap, and tie-break rules. ★ indicates unanimous agreement between both models.

Week	XGBoost	XGB Conf.	RF	RF Conf.	Final Label	Confidence	Tier	Agreed
Week 1	Moderate	86.8%	Moderate	58.9%	Moderate	72.8%	MEDIUM	★ Yes
Week 2	Moderate	62.2%	Moderate	43.3%	Moderate	52.8%	LOW	★ Yes
Week 3	Stable	74.8%	Moderate	37.1%	Stable	74.8%	MEDIUM	No
Week 4	Unstable	46.9%	Moderate	49.1%	Unstable	46.9%	LOW	No

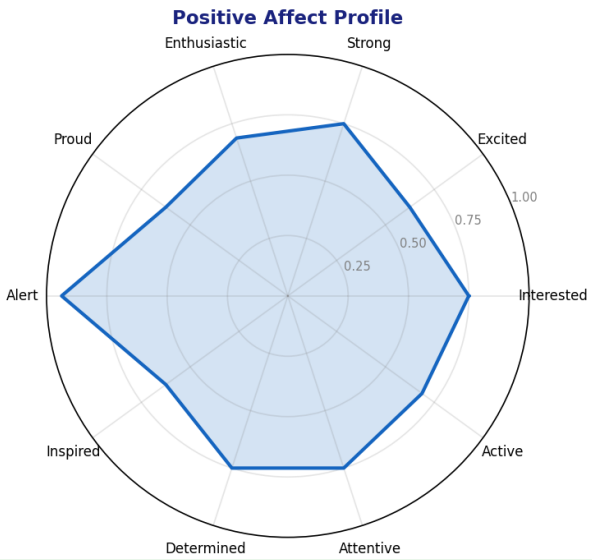




4. Emotional Dimension Analysis

Positive Affect Dimensions

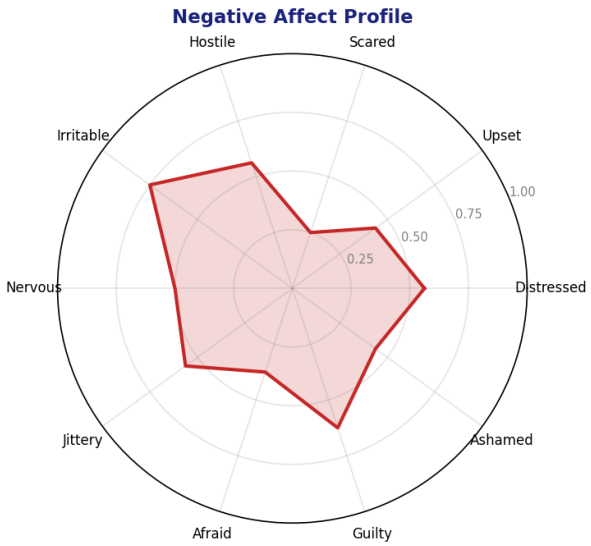
Item	Avg Score
Interested	0.750
Excited	0.625
Strong	0.750
Enthusiastic	0.688
Proud	0.625
Alert	0.938
Inspired	0.625
Determined	0.750
Attentive	0.750
Active	0.688



Positive affect showed a decreasing trend across the period. Highest-scoring dimensions: Alert and Interested.

Negative Affect Dimensions

Item	Avg Score
Distressed	0.562
Upset	0.438
Scared	0.250
Hostile	0.562
Irritable	0.750
Nervous	0.500
Jittery	0.562
Afraid	0.375
Guilty	0.625
Ashamed	0.438



Anxiety-related indicators decreased after Week 2. Dominant negative drivers: Irritable and Guilty.

5. Cognitive and Context Indicators

Indicator	Average Score
Emotional Clarity	0.500
Social Connection	0.438

Cognitive indicators were within normal range.

6. Lifestyle and Wellness Factors

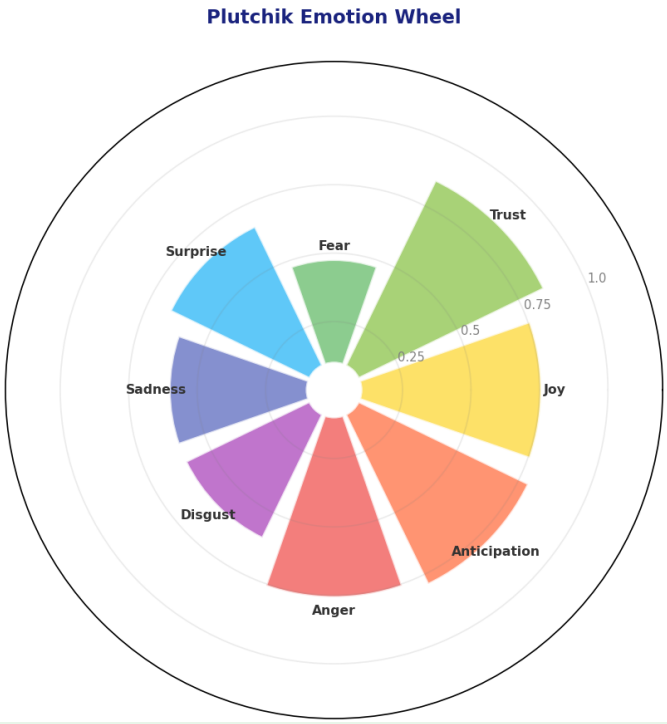
Factor	Average Score
Sleep Quality	0.562
Stress Level	0.375

Focus Ability	0.875
Physical Fatigue	0.625

Sleep quality was the dominant lifestyle factor across the reporting period.

7. Emotional Profile Visualisation

The Plutchik Wheel shows the relative strength of eight primary emotions averaged across the four-week reporting period. Petal length corresponds to mean intensity on the normalised [0,1] scale.



The wheel highlights dominant emotions and overall emotional balance over time.

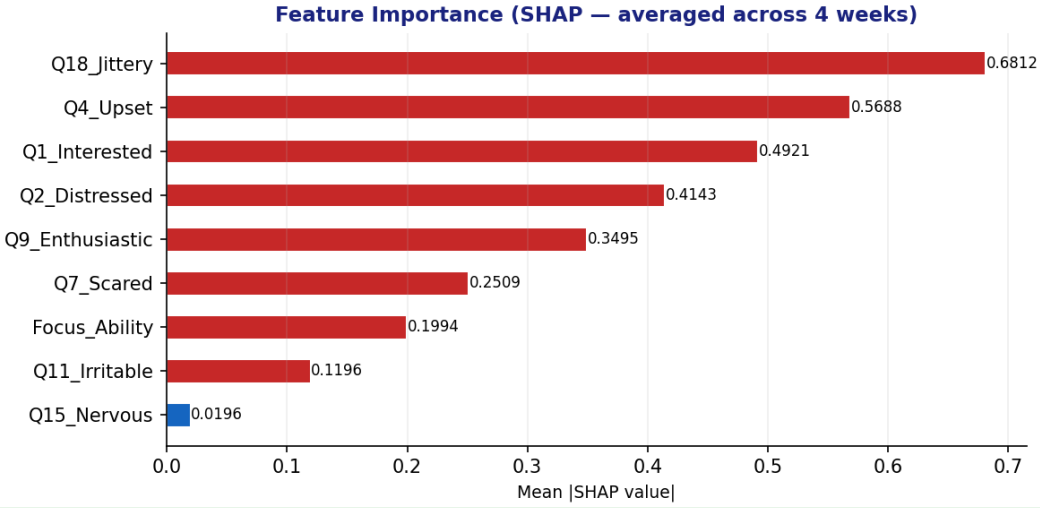
8. Mood Stability Analysis

Mood Variability Index	0.2611
Emotional Stability Score	0.25
Label Distribution	Stable × 1 | Moderate × 2 | Unstable × 1

Low variability — mood remained stable across the period.

9. Model Insights (SHAP Feature Importance)

SHAP (SHapley Additive exPlanations) values quantify each feature's contribution to the XGBoost prediction. Values shown are mean absolute SHAP importance averaged across all four weeks.



Q18_Jittery, Q4_Upset, and Q1_Interested were the strongest predictors of mood classification. In Week 4, Q4_Upset was the primary driver of the Unstable classification.

10. Observed Patterns

The following patterns were identified from your four weeks of data. Each point is drawn directly from the numbers — not generic advice.

• Your positive emotions (Positive Affect) declined across the period — from 72% in Week 1 down to 52% in Week 4. This downward trend may reflect accumulating stress or fatigue.

• Feelings of distress, anxiety, and irritability (Negative Affect) eased as the weeks progressed — from 69% in Week 1 to 62% in Week 4. Reducing negative affect, even gradually, has a meaningful positive effect on overall mood stability.

• Week 2 was your strongest week emotionally (mood balance: +0.55), while Week 4 was the most challenging (mood balance: -0.10). Understanding what was different in Week 2 — whether it was better sleep, lower stress, or more social time — could help you replicate those conditions.

• Physical fatigue was notable at 62% on average. Fatigue and emotional well-being are closely linked — persistent tiredness reduces your capacity to regulate emotions and can make challenges feel more overwhelming than they actually are.

• Among all negative emotions measured, 'Irritable' was the most prominent (avg 75%). This specific emotion — not just general negativity — was the primary contributor to your negative affect score across the period and is worth paying particular attention to.

• Your strongest positive emotion was 'Alert' (avg 94%). Positive emotions like this are not just pleasant — they actively build psychological resilience and help buffer against the effects of stress.

11. Final Summary

OVERVIEW

Over the four-week period, your emotional state progressed from Moderate in Week 1 to Unstable by Week 4 — a shift that warrants closer attention. On average, your positive emotions (energy, enthusiasm, motivation) were at 72% of their maximum, while negative emotions (stress, anxiety, irritability) were at 50% — giving an overall mood balance of +0.22. This positive balance means your emotional resources were, on average, outweighing your emotional burdens.

WHAT CHANGED ACROSS THE WEEKS

Your positive emotions declined across the period (from 72% to 52%), while negative emotions eased (from 69% to 62%). Your emotionally strongest week was Week 2 (mood balance: +0.55), and your most challenging was Week 4 (mood balance: -0.10). The variability index was 0.26 — low variability, meaning your mood was consistent week to week.

WHAT THE MODEL DETECTED

The AI model analysed your responses each week and classified your mood stability as: Stable × 1 | Moderate × 2 | Unstable × 1. The two models (XGBoost and Random Forest) agreed unanimously in 2 out of 4 weeks. The features that most influenced these classifications were: Jittery, Upset, and Interested. In plain terms, how jittery you reported feeling each week was the single strongest predictor of whether you would be classified as Stable, Moderate, or Unstable.

LIFESTYLE AND MENTAL CONTEXT

Your lifestyle data highlights several factors that shaped your mood this period: significant physical fatigue (62%). Of these, focus ability was the most prominent (difficulty concentrating often co-occurs with emotional strain). Emotional clarity averaged 50%, which is an area for growth — being able to name your feelings is the first step to managing them.

WHAT TO DO NEXT

The decline observed across the period signals that something needs to change. Start with your most actionable lever: focus ability — difficulty concentrating often co-occurs with emotional strain. If you are experiencing persistent low mood, speaking with a mental health professional is a practical and effective next step — not a sign of weakness.

IMPORTANT NOTE

This report was generated by an automated mood stability model trained on population-level data. It is designed for personal wellness awareness and self-reflection only. The classifications and interpretations in this report do not constitute a clinical diagnosis, psychological assessment, or medical advice. If you are experiencing persistent low mood, significant anxiety, or any mental health concerns, please consult a qualified mental health professional.